

Practical No. 1: Introduction to Business Analytics

- **What is Business Analytics (BA)?**

ANS– Business Analytics (BA) is the process of collecting and analyzing business data to make better business decisions.

- **What is Exploratory Data Analysis (EDA)?**

ANS–Exploratory Data Analysis (EDA) is the process of exploring and understanding a dataset before further analysis

- **Exploratory Data Analysis (EDA)**

Exploratory Data Analysis (EDA) is the first step in the data science process, where data is explored, cleaned, and visualised to gain insights and ensure its quality for further analysis and modelling.

Libraries

- **Pandas:** Used to read and analyze data.
- **Seaborn:** Used to create statistical graphs.
- **Matplotlib:** Used to create and display charts.

Functions

- `pd.read_csv()` – Loads a CSV file into a DataFrame.
- `df.info()` – Shows basic information about the dataset.
- `df.head()` – Displays the first 5 rows of the dataset.
- `df.isnull().sum()` – Checks for missing values.
- `sns.countplot()` – Creates a count graph for categories.
- `plt.title()` – Adds a title to the graph.
- `plt.show()` – Displays the graph.
- `plt.figure()` – Sets the size of the graph.
- `df.groupby()` – Groups data based on a column.
- `sum()` – Calculates the total of values.
- `plot(kind='bar')` – Creates a bar chart.
- `plt.ylabel()` – Sets the Y-axis label.

- `pd.to_datetime()` – Converts data into date format.
- `dt.month_name()` – Extracts the month name from a date.
- `sns.histplot()` – Creates a histogram of data.
- `df.corr()` – Calculates the correlation between numeric columns.
- `sns.heatmap()` – Displays correlation in a heatmap.

1. Customer Gender Distribution (Gender Count Plot)

Observation:

- The graph shows the number of male and female customers.
- Both genders have almost equal customer counts.

Analysis:

- The store is popular among both male and female customers.
 - No gender dominates the customer base.
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2. Customer Type (Member vs Normal)

Observation:

- The graph compares Member and Normal customers.
- Both customer types are present in good numbers.

Analysis:

- The store serves both regular and new customers.
 - Membership programs help in customer retention.
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3. Payment Method Used

Observation:

- The graph shows different payment methods used by customers.
- Customers use multiple payment options.

Analysis:

- Offering different payment methods improves customer convenience.
 - The most used payment method is preferred by customers.
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4. Product Line Sales

Observation:

- The graph shows the number of products sold in each product line.
- Some product lines have higher sales than others.

Analysis:

- High-selling product lines are more popular.
 - Low-selling products may need better promotion.
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5. Sales by City

Observation:

- The graph compares total sales across different cities.
- Sales vary from city to city.

Analysis:

- Cities with higher sales contribute more to business revenue.
 - Low-sales cities may require better marketing strategies.
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6. Monthly Sales Trend

Observation:

- The graph shows total sales for each month.
- Some months have higher sales than others.

Analysis:

- Sales change over time due to customer demand.
 - Peak months help identify the best business season.
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7. Customer Ratings (Histogram)

Observation:

- The graph shows the distribution of customer ratings.
- Most ratings are in the higher range.

Analysis:

- Customers are generally satisfied with the products and services.
 - Higher ratings indicate good customer experience.
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8. Correlation Heatmap

Observation:

- The heatmap shows relationships between numerical columns.
- Some variables have strong correlation, while others have weak correlation.

Analysis:

- Strong correlation means two variables are closely related.
- Weak correlation means the variables have little or no relationship.